

CBSE Class 10 Mathematics — Quadratic Equations — Solutions

1. 1. Which of the following is a quadratic equation?

Answer: (C)

Solution:

1. A quadratic equation is of the form $ax^2 + bx + c = 0$ where $a \neq 0$.
2. Check each option: (A) has x^2 on both sides cancelling out, (B) x^3 term makes it cubic, (C) simplifies to $x^2 - 3x + 2 = 0$, (D) has $1/x$ which is not polynomial.

Derivation:

$$\begin{aligned}x(x+1) + 8 &= (x+2)(x-2) \\ \implies x^2 + x + 8 &= x^2 - 4 \\ \implies x + 12 &= 0 \text{ (Linear). } (x+2)^2 = 2(x+3) \\ \implies x^2 + 4x + 4 &= 2x + 6 \\ \implies x^2 + 2x - 2 &= 0\end{aligned}$$

Quadratic Equations — Introduction to Quadratic Equations

2. 2. The discriminant of the quadratic equation $2x^2 - 4x + 3 = 0$ is:

Answer: (B)

Solution:

1. Identify $a = 2, b = -4, c = 3$.
2. Use the formula $D = b^2 - 4ac$.
3. Calculate $D = (-4)^2 - 4(2)(3) = 16 - 24 = -8$.

Derivation:

$$D = b^2 - 4ac = (-4)^2 - 4(2)(3) = 16 - 24 = -8$$

Quadratic Equations — Nature of Roots

3. 3. If $1/2$ is a root of the equation $x^2 - kx - 5/4 = 0$, then the value of k is:

Answer: (D)

Solution:

1. Substitute $x = 1/2$ into the equation.
2. $(1/2)^2 - k(1/2) - 5/4 = 0$.
3. $1/4 - k/2 - 5/4 = 0 \implies -4/4 - k/2 = 0 \implies -1 = k/2 \implies k = -2$.

Derivation:

$$\begin{aligned}\left(\frac{1}{2}\right)^2 - k\left(\frac{1}{2}\right) - \frac{5}{4} &= 0 \\ \implies \frac{1}{4} - \frac{k}{2} - \frac{5}{4} &= 0 \\ \implies -1 - \frac{k}{2} &= 0 \\ \implies k &= -2\end{aligned}$$

Quadratic Equations — Roots of a Quadratic Equation

4. **4.** The quadratic equation $x^2 + kx + 1 = 0$ has equal roots if:

Answer: (A)

Solution:

1. For equal roots, discriminant $D = 0$.
2. $b^2 - 4ac = 0$.
3. $k^2 - 4(1)(1) = 0 \implies k^2 = 4 \implies k = \pm 2$.

Derivation:

$$\begin{aligned}D &= k^2 - 4(1)(1) = 0 \\ \implies k^2 &= 4 \\ \implies k &= \pm 2\end{aligned}$$

Quadratic Equations — Nature of Roots

5. **5.** The roots of the quadratic equation $x^2 - 9 = 0$ are:

Answer: (C)

Solution:

1. Factor the equation: $x^2 - 3^2 = 0$.
2. $(x - 3)(x + 3) = 0$.
3. Therefore, $x = 3$ or $x = -3$.

Derivation:

$$\begin{aligned}x^2 &= 9 \\ \implies x &= \pm\sqrt{9} \\ \implies x &= \pm 3\end{aligned}$$

Quadratic Equations — Solving Quadratic Equations by Factorization

6. **6.** Assertion (A): The quadratic equation $x^2 + 5x + 6 = 0$ has two distinct real roots. Reason (R): A quadratic equation $ax^2 + bx + c = 0$ has distinct real roots if its discriminant $D = b^2 - 4ac > 0$.

Answer: (A)

Solution:

1. Calculate the discriminant D.
2. Check if $D \neq 0$.

Derivation:

$$D = b^2 - 4ac = (5)^2 - 4(1)(6) = 25 - 24 = 1. \text{ Since } 1 > 0, \text{ the roots are distinct and real.}$$

Quadratic Equations — Nature of Roots

7. **7.** Assertion (A): If $x = 2$ is a root of $x^2 - kx + 4 = 0$, then $k = 4$. Reason (R): A value $x = \alpha$ is a root of the equation $f(x) = 0$ if $f(\alpha) = 0$.

Answer: (A)

Solution:

1. Substitute $x = 2$ into the equation.
2. Solve for k .

Derivation:

$$\begin{aligned}(2)^2 - k(2) + 4 &= 0 \\ \implies 4 - 2k + 4 &= 0 \\ \implies 8 &= 2k \\ \implies k &= 4.\end{aligned}$$

Quadratic Equations — Roots of Quadratic Equation

8. **8.** Assertion (A): The equation $(x - 2)^2 + 1 = 2x - 3$ is a quadratic equation. Reason (R): Any equation of the form $ax^2 + bx + c = 0$ where $a \neq 0$ is a quadratic equation.

Answer: (A)

Solution:

1. Expand the LHS.
2. Simplify the equation to standard form.
3. Check if the highest power is 2.

Derivation:

$$\begin{aligned}x^2 - 4x + 4 + 1 &= 2x - 3 \\ \implies x^2 - 6x + 8 &= 0. \text{ This is in the form } ax^2 + bx + c = 0 \text{ with } a = 1.\end{aligned}$$

Quadratic Equations — Introduction to Quadratic Equations

9. **9.** Assertion (A): The quadratic equation $x^2 + 2x + 5 = 0$ has no real roots. Reason (R): For $x^2 + 2x + 5 = 0$, the discriminant is -16 .

Answer: (A)

Solution:

1. Calculate discriminant D .
2. Evaluate nature of roots based on D .

Derivation:

$$D = b^2 - 4ac = (2)^2 - 4(1)(5) = 4 - 20 = -16. \text{ Since } D < 0, \text{ there are no real roots.}$$

Quadratic Equations — Nature of Roots

10. 10. Assertion (A): The roots of $x^2 - 9 = 0$ are 3 and -3 . Reason (R): $x^2 - a^2 = (x - a)(x + a)$.

Answer: (A)

Solution:

1. Use factorisation method.
2. Solve for x.

Derivation:

$$x^2 - 9 = 0$$

$$\implies (x - 3)(x + 3) = 0. \text{ So, } x = 3 \text{ or } x = -3.$$

Quadratic Equations — Solving Quadratic Equations by Factorisation

11. 11. Find the roots of the quadratic equation $x^2 - 3x - 10 = 0$ by the method of factorization.

Answer:

$$x = 5, x = -2$$

Solution:

1. Split the middle term $-3x$ into $-5x + 2x$ such that their product is $-10x^2$.
2. Factorize the expression by grouping terms and solve for x.

Derivation:

$$x^2 - 5x + 2x - 10 = 0$$

$$\implies x(x - 5) + 2(x - 5) = 0$$

$$\implies (x - 5)(x + 2) = 0$$

$$\implies x = 5, -2$$

Quadratic Equations — Factorization Method

12. 12. Find the value of k for which the quadratic equation $2x^2 + kx + 3 = 0$ has two equal real roots.

Answer:

$$k = \pm 2\sqrt{6}$$

Solution:

1. Use the condition for equal roots, which is the discriminant $D = b^2 - 4ac = 0$.
2. Substitute $a = 2, b = k, c = 3$ into the formula and solve for k .

Derivation:

$$D = k^2 - 4(2)(3) = 0$$

$$\implies k^2 - 24 = 0$$

$$\implies k^2 = 24$$

$$\implies k = \pm\sqrt{24} = \pm 2\sqrt{6}$$

Quadratic Equations — Nature of Roots

13. 13. Check whether $x(x + 3) + 1 = (x - 2)(x + 2)$ is a quadratic equation.

Answer:

No, it is not a quadratic equation.

Solution:

1. Expand both sides of the equation to simplify.
2. Check if the resulting equation is of the form $ax^2 + bx + c = 0$ where $a \neq 0$.

Derivation:

$$\begin{aligned}x^2 + 3x + 1 &= x^2 - 4 \\ \implies 3x + 1 &= -4 \\ \implies 3x + 5 &= 0. \text{ Since the degree is 1, it is linear, not quadratic.}\end{aligned}$$

Quadratic Equations — Introduction to Quadratic Equations

14. 14. Find the roots of the equation $2x^2 - x + \frac{1}{8} = 0$ by using the quadratic formula.

Answer:

$$x = \frac{1}{4}$$

Solution:

1. Multiply the entire equation by 8 to simplify the coefficients.
2. Apply the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

Derivation:

$$16x^2 - 8x + 1 = 0. \text{ Here } a = 16, b = -8, c = 1.$$

$$\implies x = \frac{8 \pm \sqrt{64 - 64}}{32} = \frac{8}{32} = \frac{1}{4}$$

Quadratic Equations — Quadratic Formula

15. 15. The sum of two numbers is 27 and their product is 182. Form the quadratic equation representing this scenario.

Answer:

$$x^2 - 27x + 182 = 0$$

Solution:

1. Let one number be x , then the other number is $27 - x$.
2. Set up the equation based on their product $x(27 - x) = 182$ and simplify.

Derivation:

$$\begin{aligned}x(27 - x) &= 182 \\ \implies 27x - x^2 &= 182 \\ \implies x^2 - 27x + 182 &= 0\end{aligned}$$

Quadratic Equations — Word Problems

16. **16.** Find the roots of the quadratic equation $2x^2 - x + \frac{1}{8} = 0$ by the method of factorization.

Answer:

$$x = \frac{1}{4}, \frac{1}{4}$$

Solution:

1. Multiply the entire equation by 8 to eliminate the fraction: $16x^2 - 8x + 1 = 0$.
2. Split the middle term: $16x^2 - 4x - 4x + 1 = 0$.
3. Factor by grouping: $4x(4x - 1) - 1(4x - 1) = 0$.
4. Solve for x: $(4x - 1)^2 = 0$, thus $x = 1/4$.

Derivation:

$$\begin{aligned}16x^2 - 8x + 1 &= 0 \\16x^2 - 4x - 4x + 1 &= 0 \\4x(4x - 1) - 1(4x - 1) &= 0 \\(4x - 1)(4x - 1) &= 0 \\x &= 1/4\end{aligned}$$

Quadratic Equations — Factorization Method

17. **17.** Find two consecutive positive integers, sum of whose squares is 365.

Answer:

$$13 \text{ and } 14$$

Solution:

1. Let the two consecutive integers be x and $x + 1$.
2. Set up the equation: $x^2 + (x + 1)^2 = 365$.
3. Expand and simplify to standard form: $2x^2 + 2x - 364 = 0$, or $x^2 + x - 182 = 0$.
4. Solve using factorization: $(x + 14)(x - 13) = 0$. Since integers are positive, $x = 13$.

Derivation:

$$\begin{aligned}x^2 + x^2 + 2x + 1 &= 365 \\2x^2 + 2x - 364 &= 0 \\x^2 + x - 182 &= 0 \\(x + 14)(x - 13) &= 0 \\x &= 13, x + 1 = 14\end{aligned}$$

Quadratic Equations — Word Problems

18. **18.** Find the value of k for which the quadratic equation $2x^2 + kx + 3 = 0$ has two equal real roots.

Answer:

$$k = \pm 2\sqrt{6}$$

Solution:

1. For equal real roots, the discriminant $D = b^2 - 4ac$ must be equal to 0.
2. Identify coefficients: $a = 2, b = k, c = 3$.
3. Substitute into discriminant formula: $k^2 - 4(2)(3) = 0$.
4. Solve for k : $k^2 - 24 = 0$, so $k^2 = 24$, $k = \pm\sqrt{24} = \pm 2\sqrt{6}$.

Derivation:

$$\begin{aligned} b^2 - 4ac &= 0 \\ k^2 - 4(2)(3) &= 0 \\ k^2 &= 24 \\ k &= \pm 2\sqrt{6} \end{aligned}$$

Quadratic Equations — Nature of Roots

- 19. 19.** The altitude of a right triangle is 7 cm less than its base. If the hypotenuse is 13 cm, find the other two sides.

Answer:

$$\text{Base} = 12\text{cm}, \text{Altitude} = 5\text{cm}$$

Solution:

1. Let base be x cm, then altitude is $(x - 7)$ cm.
2. Apply Pythagoras theorem: $x^2 + (x - 7)^2 = 13^2$.
3. Expand and simplify: $x^2 + x^2 - 14x + 49 = 169 \Rightarrow 2x^2 - 14x - 120 = 0$.
4. Divide by 2: $x^2 - 7x - 60 = 0$. Factorize: $(x - 12)(x + 5) = 0$. Base must be positive, so $x = 12$.

Derivation:

$$\begin{aligned} x^2 + (x - 7)^2 &= 13^2 \\ 2x^2 - 14x + 49 - 169 &= 0 \\ 2x^2 - 14x - 120 &= 0 \\ x^2 - 7x - 60 &= 0 \\ (x - 12)(x + 5) &= 0 \end{aligned}$$

Quadratic Equations — Word Problems

- 20. 20.** Solve the quadratic equation $x^2 - 5x + 6 = 0$ using the quadratic formula.

Answer:

$$x = 2, 3$$

Solution:

1. Identify coefficients: $a = 1, b = -5, c = 6$.
2. Calculate the discriminant: $D = b^2 - 4ac = (-5)^2 - 4(1)(6) = 25 - 24 = 1$.
3. Use the quadratic formula $x = \frac{-b \pm \sqrt{D}}{2a}$.
4. Substitute values: $x = \frac{5 \pm 1}{2}$. This gives $x = \frac{6}{2} = 3$ and $x = \frac{4}{2} = 2$.

Derivation:

$$\begin{aligned}x &= \frac{-(-5) \pm \sqrt{(-5)^2 - 4(1)(6)}}{2(1)} \\x &= \frac{5 \pm \sqrt{25 - 24}}{2} \\x &= \frac{5 \pm 1}{2} \\x &= 3, 2\end{aligned}$$

Quadratic Equations — Quadratic Formula

- 21. 21.** A motorboat whose speed is 18 km/h in still water takes 1 hour more to go 24 km upstream than to return downstream to the same spot. Find the speed of the stream.

Answer:

$$6\text{km/h}$$

Solution:

1. Let the speed of the stream be x km/h.
2. Speed upstream = $(18 - x)$ km/h; Speed downstream = $(18 + x)$ km/h.
3. Time taken upstream $T_1 = \frac{24}{18-x}$ and time taken downstream $T_2 = \frac{24}{18+x}$.
4. Set up the equation: $T_1 - T_2 = 1$.
5. Solve the resulting quadratic equation $x^2 + 48x - 324 = 0$.
6. Factorize to find $x = 6$ or $x = -54$. Since speed cannot be negative, $x = 6$.

Derivation:

$$\begin{aligned}\frac{24}{18-x} - \frac{24}{18+x} &= 1 \\ \implies 24(18+x-18+x) &= (18-x)(18+x) \\ \implies 48x &= 324 - x^2 \\ \implies x^2 + 48x - 324 &= 0 \\ \implies (x+54)(x-6) &= 0\end{aligned}$$

Quadratic Equations — Word problems involving speed, distance, and time

- 22. 22.** The sum of the areas of two squares is 468 m². If the difference of their perimeters is 24 m, find the sides of the two squares.

Answer:

$$12\text{m and } 18\text{m}$$

Solution:

1. Let sides of squares be x and y ($x > y$).
2. Perimeters are $4x$ and $4y$. Given $4x - 4y = 24$, so $x - y = 6 \implies x = y + 6$.
3. Sum of areas: $x^2 + y^2 = 468$.
4. Substitute x : $(y + 6)^2 + y^2 = 468$.
5. Expand and simplify: $y^2 + 12y + 36 + y^2 = 468 \implies 2y^2 + 12y - 432 = 0$.

6. Simplify to $y^2 + 6y - 216 = 0$ and solve for y using the quadratic formula or factoring.

Derivation:

$$\begin{aligned} y^2 + 6y - 216 &= 0 \\ \implies (y + 18)(y - 12) &= 0 \\ \implies y = 12 \text{ (since side } > 0), x = 12 + 6 = 18 \end{aligned}$$

Quadratic Equations — Geometric problems

- 23. 23.** A peacock is sitting on the top of a pillar, which is 9 m high. From a point 27 m away from the bottom of the pillar, a snake is coming to its hole at the base of the pillar. Seeing the snake, the peacock pounces on it. If their speeds are equal, at what distance from the hole is the snake caught?

Answer:

$$12m$$

Solution:

1. Let the snake be caught at distance x from the hole. The snake travels $27 - x$ distance.
2. The peacock flies from the top of the pillar to the point of catch. By Pythagoras theorem, distance covered by peacock $= \sqrt{9^2 + x^2}$.
3. Since speeds are equal and they start at the same time, distances are equal: $27 - x = \sqrt{81 + x^2}$.
4. Square both sides: $(27 - x)^2 = 81 + x^2$.
5. Expand: $729 - 54x + x^2 = 81 + x^2$.
6. Solve for x : $54x = 648 \implies x = 12$.

Derivation:

$$\begin{aligned} 729 - 54x &= 81 \\ \implies 54x &= 648 \\ \implies x &= \frac{648}{54} = 12 \end{aligned}$$

Quadratic Equations — Application of quadratic equations in geometry

- 24. 24.** A shopkeeper buys a number of books for 80 rupees. If he had bought 4 more books for the same amount, each book would have cost 1 rupee less. How many books did he buy?

Answer:

$$16books$$

Solution:

1. Let the number of books be x .
2. Original cost per book $= \frac{80}{x}$.
3. New number of books $= x + 4$. New cost per book $= \frac{80}{x+4}$.
4. Given the difference in price is 1: $\frac{80}{x} - \frac{80}{x+4} = 1$.

5. Simplify: $80(x + 4) - 80x = x(x + 4)$.
6. Form quadratic: $x^2 + 4x - 320 = 0$.
7. Solve: $(x + 20)(x - 16) = 0$. Since $x > 0$, $x = 16$.

Derivation:

$$\begin{aligned}
 80x + 320 - 80x &= x^2 + 4x \\
 \implies x^2 + 4x - 320 &= 0 \\
 \implies x &= \frac{-4 \pm \sqrt{16 - 4(1)(-320)}}{2} = \frac{-4 \pm 36}{2}
 \end{aligned}$$

Quadratic Equations — Word problems based on commercial mathematics